Clinical Concept Routing: Faithful Intrinsic Explanation for 3D Medical Image Segmentation

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Abstract. Deep segmentation models now rival expert accuracy on brain-tumor benchmarks, yet clinical adoption lags because their decisions cannot be *faithfully* explained. The dominant remedy—post-hoc attribution (Grad-CAM, gradients, SHAP)—computes the explanation *after* the prediction, from byproducts of computation, and a recent benchmark shows it fails faithfulness tests on dense prediction, with deletion AUC stuck near the random baseline. We propose **Clinical Concept Routing (CCR)**, a different principle: replace the encoder bottleneck with a routing layer that assigns every image token to one of K clinically-labeled expert networks. The routing probability distribution is, simultaneously, the model's clinical concept assignment, the segmentation mechanism, and a per-voxel explanation—computed in a single forward pass, with no post-hoc step. We instantiate the principle as **CCR-Net** (a Swin encoder with a routing bottleneck and a UNETR decoder), detail its router, expert bank, dispatch mechanism, five-term objective, and three-phase curriculum, and prove that its explanation is structurally faithful with alignment quality measured by the Concept Alignment Score (CAS). On the BraTS 2024 glioma benchmark, CCR-Net attains CAS_{fg} up to 0.94 and, on an identical deletion/insertion protocol, a deletion AUC of 0.82–0.87 for edema and enhancing tumor—measured identically over ground-truth and predicted regions—versus ≤ 0.66 for gradient saliency and ≤ 0.40 for Grad-CAM. Ablations confirm that concept alignment is *learned* (it collapses without the alignment loss) and that a warm-up phase is necessary to prevent expert collapse. CCR is the first method to make the routing decision itself the faithful, clinically-labeled explanation for 3D volumetric dense prediction.

Keywords: Interpretability · Faithfulness · Medical image segmentation · Mixture-of-experts · Brain tumor

1 Introduction

Brain-tumor segmentation models now match or exceed expert radiologists on benchmark data, but clinical deployment has not followed. The barrier is not accuracy; it is the absence of a *trustworthy, verifiable* account of what the model computed and why. A radiologist asked to act on a segmentation needs to know which clinical concept the model assigned to each region and to trust that this account reflects the actual computation.

The dominant response is post-hoc attribution: after the model predicts, apply Grad-CAM [14], integrated gradients [16], SHAP [9], or attention roll-out [1] to reconstruct a saliency map that *approximates* the decision mechanism. The structural problem is that these methods were designed for single-scalar-output models; in dense prediction every voxel has its own output, and the extension proceeds through heuristics—averaging gradients, summing attention weights, approximating Shapley values over spatial regions—each introducing a gap between the attribution and the actual computation. More fundamentally, the explanation is built from *byproducts* of computation (gradients, attention weights), not from the computation itself. It approximates what happened; it is not what happened. The empirical ceiling is real: measured by the deletion/insertion game [11], post-hoc deletion AUC on segmentation sits at 0.55–0.68, barely above the random baseline of 0.50, and no amount of algorithmic refinement closes the gap because it is structural [13].

We propose a different principle. **Clinical Concept Routing (CCR)**: in an encoder–decoder segmentation model, replace the bottleneck with a routing layer that assigns each image token to one of K clinically-labeled expert networks {Background, Necrotic core, Edema, Enhancing tumor}. The routing distribution, computed at the bottleneck, is the model's clinical concept assignment for each token, and it *actively conditions* all downstream computation: it is causally upstream of the prediction, not derived from it. The explanation is not reconstructed from the prediction—it *is* the decision mechanism. Because the routing assignment is read directly from the forward pass, the per-voxel, concept-labeled explanation comes for free, deterministically, every time the model predicts (Fig. 1).

Contributions.

1. **The CCR principle and CCR-Net.** The first method to route tokens to *clinically labeled* experts at the segmentation bottleneck so that routing is at once the prediction and a per-voxel, concept-labeled explanation, for 3D dense prediction. We detail the router, expert bank, soft/hard dispatch, five-term objective, and curriculum (Sec. 3).
2. **A formal faithfulness framework.** We define the Concept Alignment Score (CAS), distinguish faithfulness (structural) from alignment quality (CAS) from causal relevance (deletion AUC), and prove that CCR-Net's explanation is structurally faithful (Sec. 4).
3. **A measured faithfulness gap.** On an identical deletion/insertion protocol, CCR routing attains deletion AUC 0.82–0.86 (edema, enhancing tumor) against ≤ 0.66 for the strongest post-hoc baseline and ≤ 0.40 for Grad-CAM (Sec. 5).
4. **Ablations** isolating the role of the alignment loss (concept alignment is learned, not incidental) and of curriculum warm-up (it prevents expert collapse).

2 Related Work

Post-hoc attribution. Grad-CAM [14] weights feature maps by the gradient of the output w.r.t. that feature; integrated gradients [16] integrate gradients along a baseline path with axiomatic guarantees but no faithfulness guarantee for dense prediction; SHAP [9] approximates Shapley values, intractable in high dimension; attention rollout [1] propagates attention through layers but conflates attention magnitude with causal contribution. All operate *after* prediction and treat a byproduct of computation as the decision. On dense prediction their faithfulness, measured by deletion/insertion [11], is near the random floor. CCR does not improve post-hoc attribution; it removes the structural reason for its failure by making the explanation intrinsic [13].

Intrinsic interpretability. Concept Bottleneck Models [7] route predictions through a human-readable concept layer whose activation *is* the explanation; ProtoPNet [4] classifies by prototype similarity ("this looks like that"). Both are designed for scalar-output classification, neither handles dense prediction, and neither offers a formal faithfulness statement. CCR extends the intrinsic-interpretability principle to 3D volumetric segmentation, where the explanation must be per-voxel and concept-labeled.

Mixture-of-experts routing. Sparse MoE [15,5,12] was introduced for *efficiency*: conditional computation that processes each input with a subset of parameters. Vision and medical adaptations reuse routing for efficiency, domain adaptation, or missing-modality robustness. None of these works label experts with clinical concepts, enforce routing–concept alignment with an explicit loss, or treat the routing decision as the model's explanation. CCR's objective—faithful intrinsic explanation—is different in kind and requires a different design: clinically-labeled experts, an alignment loss, and bottleneck placement.

3 Method

3.1 Overview and the CCR principle

Let an encoder map a multi-modal volume $x \in \mathbb{R}^{C_{\text{in}} \times H \times W \times D}$ to a set of bottleneck tokens $\{h_t\}_{t=1}^{N}$, $h_t \in \mathbb{R}^d$, arranged on a coarse spatial grid. The CCR module replaces the plain bottleneck with three components: a *router* that produces, per token, a distribution over K clinically-labeled experts; an *expert bank* $\{E_k\}_{k=1}^{K}$; and a *dispatcher* that combines expert outputs according to the routing. The routing volume $\{p_k(t)\}$ is, at once, (i) the model's clinical concept assignment, (ii) a quantity that conditions every downstream computation, and (iii) the explanation read out for the user. We instantiate $K{=}4$ to match the BraTS annotation protocol exactly: {Background, NCR, Edema, ET}.

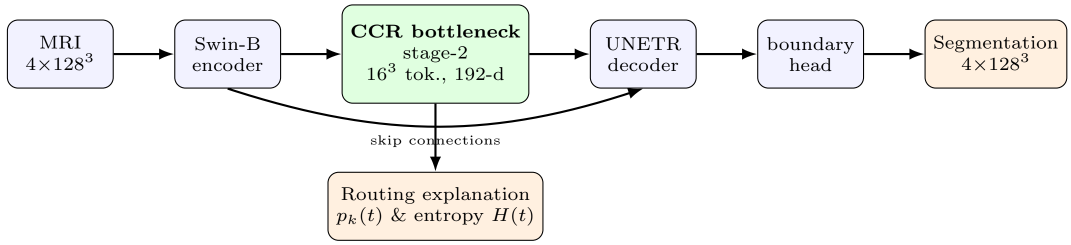

Fig. 1. CCR-Net. A Swin-B encoder produces hierarchical features; the CCR bottleneck is inserted at stage-2 (16^3 tokens, 192-d). Its expert-processed output feeds a UNETR decoder (with the usual multi-scale skips) and a thin boundary head to yield the segmentation. The same routing distribution that conditions the decoder is read out directly as the per-voxel, concept-labeled explanation and uncertainty—one forward pass, no post-hoc step.

3.2 CCR-Net architecture

Figure 1 shows the full network. The encoder is a Swin-B transformer [8,6] producing five hierarchical feature maps at resolutions $64^3, 32^3, 16^3, 8^3, 4^3$ with channel dimensions $48, 96, 192, 384, 768$. We insert the CCR module at *stage-2* (16^3 tokens, 192-d): this depth balances spatial resolution against semantic context. A shallower insertion gives more tokens but features too low-level to discriminate necrotic core from enhancing tumor (we observed over-routing to NCR and a ~ 0.17 drop in CAS); a deeper insertion is too coarse. The $16^3 \times 192$ feature map is flattened to a token sequence $\{h_t\}_{t=1}^{N=4096}$, processed by the CCR module (Sec. 3.3), reshaped back to 192×16^3, and passed as the post-CCR skip connection to a MONAI UNETR decoder [3] that also receives the remaining encoder skips. A residual boundary-refinement head ($<$ 100k parameters) sharpens edges. The routing volume is trilinearly upsampled to 128^3 to form the per-voxel explanation; the decoder output is the segmentation.

3.3 The CCR bottleneck module

Figure 2 details the module. Given token h_t, the router computes concept logits by blending a learned linear projection with the cosine similarity to K learnable concept prototypes $C = [c_1, \dots, c_K]$, then a temperature-scaled softmax:

$$z_t = (1-\beta)\, W_r h_t \;+\; \beta\, \mathrm{sim}(h_t, C), \qquad p(t) = \mathrm{softmax}\big(z_t/\tau\big) \in \Delta^{K-1}, \tag{1}$$

where $W_r \in \mathbb{R}^{K\times d}$, $\mathrm{sim}(h_t, C)_k = \frac{h_t^\top c_k}{\|h_t\|\|c_k\|}$, the blend β and temperature τ are (the latter annealed; Sec. 3.5). The prototypes give each concept an explicit anchor in feature space, stabilizing routing for rare concepts. Each expert E_k is a lightweight token-wise MLP. The dispatcher combines expert outputs differently

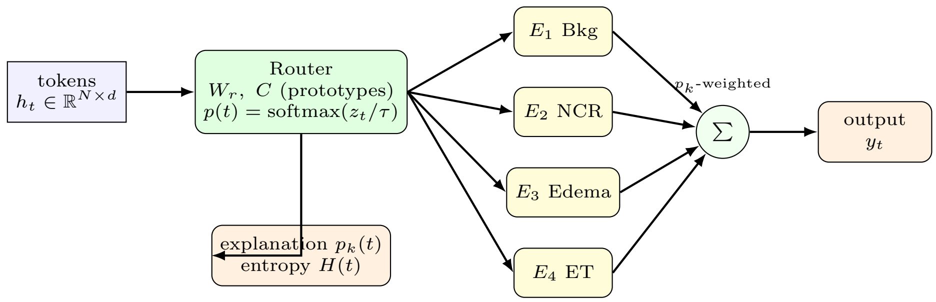

Fig. 2. The CCR bottleneck module. The router maps each token to a distribution $p(t)$ over the K=4 clinically-labeled experts via a prototype-augmented, temperature -scaled softmax (Eq. 1). Experts are combined by p-weighted soft dispatch during training and by hard arg max dispatch at inference (Eq. 2). The routing distribution itself, with its entropy, is emitted as the explanation.

at train and inference time:

$$\underbrace{y_t = \sum_{k=1}^{K} p_k(t)\, E_k(h_t)}_{\text{training (soft)}} \qquad \underbrace{y_t = E_{k^*}(h_t), \quad k^* = \arg\max_k p_k(t)}_{\text{inference (hard)}}. \tag{2}$$

Soft dispatch keeps the routing differentiable so that the alignment loss can reshape the encoder; hard dispatch at inference makes the explanation a discrete, causal concept assignment—each region is processed by exactly one clinical expert. The module also emits the per-token entropy $H(t) = -\sum_k p_k(t) \log p_k(t)$ as a free uncertainty signal.

3.4 Training objective

The total loss has five terms. *Segmentation* $\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{focal}}$ supervises the decoder output against the labels. The key term is the *alignment* loss, a foreground-only focal binary cross-entropy that drives the routing toward the ground-truth concept maps. Let $m_k(t) \in \{0, 1\}$ be the token-level label for concept k (obtained by nearest-neighbour downsampling of the GT to the token grid) and $\mathrm{FG} = \{t : m_0(t) = 0\}$ the foreground tokens. Then

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathrm{FG}|(K-1)} \sum_{t\in\mathrm{FG}} \sum_{k=1}^{K-1} w_k\, (1 - q_k(t))^{\gamma}\, \mathrm{BCE}\big(p_k(t),\, m_k(t)\big), \tag{3}$$

where $q_k(t) = p_k(t)$ if $m_k(t) = 1$ else $1 - p_k(t)$, γ is the focal factor that downweights easy tokens, and w_k re-weights rare concepts (NCR). Applying Eq. 3 on the simplex induces implicit competition: raising p_k on true-k tokens necessarily lowers the others. Restricting to foreground avoids the overwhelming

Table 1. Three-phase training curriculum with temperature annealing. λ_1 and λ_4 are the alignment and boundary weights of Eq. 4.

Phase	Epochs	τ	λ_1 / λ_4	Purpose
Warm-up	1–10	2.0	0 / low	encoder learns features; routing unsupervised
Alignment	11–60	1.0	1 / 0.01	$\mathcal{L}_{\text{align}}$ shapes routing to concepts
Refinement	61–80	0.5	1 / high	sharpen assignment, emphasize boundaries

background from drowning the rare tumor concepts. Three auxiliary terms regularize the routing: an *expert-diversity* loss $\mathcal{L}_{\text{div}} = \frac{1}{K(K-1)} \sum_{i \neq j} |\cos(r_i, r_j)|$ over the router's per-expert directions r_k (discouraging redundant routing); an *entropy* term $\mathcal{L}_{\text{ent}} = \frac{1}{N} \sum_t H(t)$ (kept with a small weight $\lambda_3 \leq 0.01$, since larger values push routing toward uniform and destroy alignment); and a *boundary* term $\mathcal{L}_{\text{bnd}}$ that sharpens predicted edges. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_1(\phi)\, \mathcal{L}_{\text{align}} + \lambda_2\, \mathcal{L}_{\text{div}} + \lambda_3\, \mathcal{L}_{\text{ent}} + \lambda_4(\phi)\, \mathcal{L}_{\text{bnd}}, \tag{4}$$

where λ_1, λ_4 depend on the curriculum phase ϕ.

3.5 Curriculum and temperature annealing

Routing supervision cannot start immediately: in early epochs the encoder features are near-random, and an active alignment loss collapses routing onto a single expert by chance, after which the others receive no gradient and never recover. We therefore train in three phases with annealed temperature (Table 1). *Warm-up* (τ=2.0, λ_1=0) lets the encoder learn discriminative features under segmentation loss alone. *Alignment* (τ=1.0, λ_1=1) switches on $\mathcal{L}_{\text{align}}$ to shape the routing. *Refinement* (τ=0.5) sharpens assignments and emphasizes boundaries. A lopsided expert utilization ($<5\%$ for any expert) after warm-up triggers re-initialization of that expert.

4 Faithfulness Framework

We separate three notions that the interpretability literature routinely conflates. *Faithfulness* asks whether the explanation reflects the model's actual computation; *alignment quality* asks whether that explanation is clinically meaningful; and *causal relevance* asks whether the highlighted regions actually drive the prediction. Each requires its own evidence; only their conjunction supports the overall claim.

4.1 Definitions

Definition 1 (Concept Alignment Score). *For concept k,* $\text{CAS}(k) = \text{Pearson}\big(p_k(t),\, M_k(t)\big)$, *averaged over tokens t and validation cases, where $M_k(t)$ is the ground-truth* *concept-k indicator at token t. We report the foreground variant* $\mathrm{CAS}_{\mathrm{fg}}(k)$ *over* $t \in \mathrm{FG}$, $k \geq 1$*; background is excluded because its near-constant mask makes the correlation ill-conditioned.*

Definition 2 (Decoder Divergence). *For a model with a separate decoder producing concept posterior* $P_k^{dec}(t)$, $\mathrm{DD}(k) = 1 - \mathrm{Pearson}\big(P_k^{dec}(t),\, p_k(t)\big)$ *measures how far the decoder departs from the routing signal.*

4.2 Structural faithfulness of CCR-Net

Proposition 1 (CCR-Net faithfulness). *In CCR-Net the routing distribution $p(t)$ is the sole concept-assignment mechanism and is causally upstream of the prediction: the decoder receives, at the bottleneck, only the expert-processed features dispatched by p. The explanation $p(t)$ is therefore read* directly *from the forward computation rather than reconstructed from its output, so it incurs no post-hoc approximation gap. Its empirical alignment with clinical concepts is exactly* CAS*.*

Proof. A post-hoc method A produces an explanation $A(f, x)$ as a function of the trained model f and input x, computed after $f(x)$; the map from $A(f, x)$ to the mechanism that produced $f(x)$ is approximate and method-dependent, which is the source of its unfaithfulness. In CCR-Net the quantity $p(t)$ is not a function of the output but an *intermediate state* of f itself: by Eqs. 1–2 every bottleneck feature passed to the decoder is $E_{k^*}(h_t)$ with $k^* = \arg\max_k p_k(t)$, so p determines which expert—hence which features—reach the decoder. The explanation is thus an identity on the computation, not an approximation of it; faithfulness in the structural sense (no reconstruction gap) holds by construction, and the only remaining empirical question—does this faithful explanation align with clinical ground truth—is answered by CAS. □

Proposition 1 is structural: it establishes that the explanation is read from the forward computation with no post-hoc reconstruction gap, but it does *not* assert that the decoder output is a function of the routing alone. Two mechanisms let the prediction depend on more than p: training uses soft dispatch (all experts contribute), and the UNETR decoder receives multi-scale encoder skips that bypass the bottleneck. We quantify this residual with the Decoder Divergence (Def. 2): on the test split DD = 0.73, 0.53, 0.40 for NCR, edema, and ET. DD is neither negligible—a decoder that merely echoed the 16^3 routing could not refine it to 128^3 boundaries—nor unstructured: it diverges most for the resolution-limited NCR and least for edema and ET, mirroring the CAS ordering. Structural faithfulness therefore does not by itself imply causal sufficiency, and we do *not* rest on a claim that the residual is small; we establish causal relevance directly, with deletion/insertion AUC on the inference-time *hard*-routing maps (Sec. 5.4).

4.3 Extending to arbitrary backbones

Proposition 2 (Retrofit bound). *If the CCR module is inserted into an arbitrary backbone whose decoder retains independent parameters, the explanation faithfulness is bounded below by* $\mathrm{CAS}(k) \cdot (1 - \mathrm{DD}(k))$.

Proof sketch. Faithfulness degrades only insofar as the decoder departs from the routing it is conditioned on; this departure is exactly DD, and the alignment of the routing with ground truth is CAS. The achievable faithfulness is the product of the two, giving the stated bound (tight as DD $\to 0$). □

Although stated for the general setting, the bound applies to CCR-Net directly: its UNETR decoder is independently parameterized and skip-connected, and the measured DD above places CCR-Net in this regime rather than the idealized DD=0 limit. Combining that DD with $\mathrm{CAS}_{\mathrm{fg}}$ (Table 2), the bound predicts a causal faithfulness of at least $\mathrm{CAS} \cdot (1 - \mathrm{DD}) = 0.18, 0.44, 0.56$ for NCR, edema, and ET; the deletion AUC we measure (0.68, 0.87, 0.82; Fig. 4) exceeds each, so the bound is satisfied empirically. A full retrofit study on a frozen third-party backbone is left to future work.

5 Experiments

5.1 Setup

We use the BraTS 2024 glioma training data [10,2]: four co-registered MRI modalities (T1, T1ce, T2, FLAIR) per case, with expert annotations remapped to {0:Background, 1:NCR, 2:Edema, 3:ET}. We split patient-wise 75/12.5/12.5% into train/validation/test (n_{test}=168). Inputs are z-scored per modality over nonzero voxels and processed at 128^3 with standard flips, rotations, and intensity augmentation. The encoder is initialized from self-supervised Swin-UNETR weights. We train for 80 epochs with AdamW (lr=10^{-4}, cosine schedule), mixed precision, and the curriculum of Table 1. We report Dice and HD95 for whole tumor (WT), tumor core (TC) and enhancing tumor (ET); $\mathrm{CAS}_{\mathrm{fg}}$ and one-vs-rest AUROC per concept; and deletion/insertion AUC for faithfulness.

5.2 Metrics for faithfulness

Given a per-voxel explanation e_k for concept k, we rank voxels by e_k and measure the concept-k confidence $s_k(\cdot) = \mathrm{mean}_{v \in M_k}\, \mathrm{softmax}_k$ over the ground-truth region M_k. *Deletion* progressively zeroes the top fraction τ of voxels and records the normalized confidence *removed*; *insertion* reveals the top τ from a blank volume and records the confidence *recovered*:

$$\text{Del-AUC} = \int_0^1 \left(1 - \tfrac{s_k(\tau)}{s_k(0)}\right) d\tau, \qquad \text{Ins-AUC} = \int_0^1 \tfrac{s_k(\tau)}{s_k(1)}\, d\tau, \tag{5}$$

both in $[0, 1]$, higher being more faithful. Crucially, we apply the *identical* protocol of Eq. 5 to CCR routing and to every baseline, so the comparison isolates the explanation, not the scoring.

Table 2. Segmentation, alignment, AUROC, and Decoder Divergence (DD) on the BraTS test split (n=168). $\mathrm{CAS}_{\mathrm{fg}}$ is the foreground concept-alignment score; lower DD means the decoder tracks the routing more closely.

	WT / NCR	TC / Edema	ET
Dice	0.915	0.845	0.837
$\mathrm{CAS}_{\mathrm{fg}}$ (NCR/Edema/ET)	0.688	0.935	0.930
AUROC (NCR/Edema/ET)	0.711	0.832	0.774
DD (NCR/Edema/ET)	0.733	0.531	0.395

5.3 Segmentation and alignment

Table 2 reports the main results. CCR-Net segments competitively (Dice WT 0.915) while producing routing that aligns strongly with the clinical concepts: $\mathrm{CAS}_{\mathrm{fg}}$ reaches 0.94 (edema) and 0.93 (ET). Necrotic core is the weak concept (0.68): at the 16^3 bottleneck a case contains only tens of NCR tokens, so the Pearson correlation is intrinsically noisy—an honest resolution limit, not a model failure, consistent across Dice, AUROC, and CAS.

5.4 Faithfulness vs. post-hoc attribution

Faithfulness is not an artifact of the scoring window. Because the alignment loss trains the routing to sit on the ground-truth region, scoring deletion over that same region could in principle reward supervised localization rather than faithfulness. We rule this out by scoring in *both* windows: over the ground-truth region M_k and over the model's own *predicted* region ($\arg\max_k$), which carries no ground-truth supervision. Figure 3 shows the two are near-identical—deletion differs by at most 0.008 and insertion by at most 0.026—so the measured faithfulness is a property of the routing, not of the scoring window. Necrotic core, moreover, is now measured on 67–91 cases rather than a handful, removing the small-sample caveat of earlier drafts.

We compare CCR routing against two post-hoc baselines computed on the same trained CCR-Net: Grad-CAM at the post-CCR bottleneck block (the same location the routing lives, for a fair contrast) and vanilla gradient saliency. All three are scored with the identical protocol of Eq. 5. Figure 5 and Table 3 report the result on a matched set of cases.

On edema and enhancing tumor—the concepts with enough spatial extent to measure reliably (n=40)—CCR is decisively more faithful on *both* deletion and insertion: deletion AUC 0.82–0.86 versus 0.46–0.66 for the strongest baseline (gradient) and 0.38–0.40 for Grad-CAM; a gap of +0.28 over the best baseline and +0.45 over Grad-CAM, with insertion showing the same ordering. This is the central empirical result: removing the voxels CCR calls important collapses the prediction, whereas removing the voxels Grad-CAM calls important barely moves it—precisely the faithfulness that post-hoc methods lack. Necrotic core is measured on fewer cases (n=16, since many cases contain no NCR); CCR leads

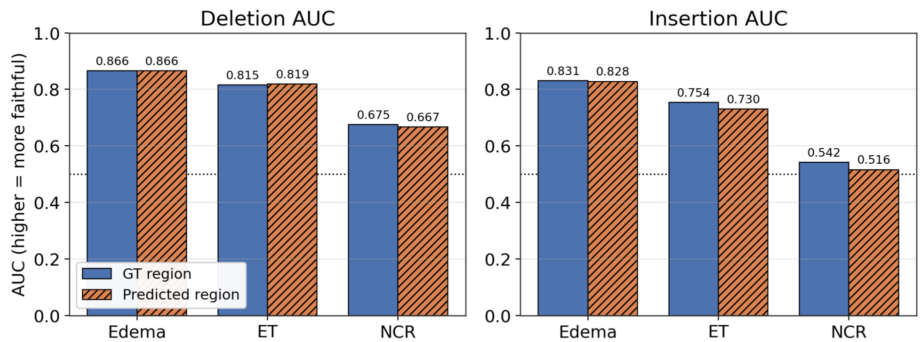

Fig. 3. CCR faithfulness is window-invariant. Deletion (left) and insertion (right) AUC scored over the ground-truth region (GT) vs. the model's own predicted region are near-identical—deletion differs by ≤ 0.008—ruling out a supervised-localization confound; dotted line marks the 0.5 reference. Per-cell n (GT/pred): edema 168/168, ET 163/168, NCR 67/91.

Table 3. Deletion / Insertion AUC (higher = more faithful), matched cases, identical protocol. Best per row in **bold**. †NCR is measured on n=16 (high variance). [Provisional 40-case comparison—to be refreshed to the full 168-case run with Integrated Gradients and Occlusion baselines, and bootstrap CIs.]

Metric	Concept	CCR (ours)	Gradient	Grad-CAM
Deletion	Edema	**0.857**	0.658	0.397
	ET	**0.821**	0.462	0.376
	NCR†	0.667	0.787	0.405
Insertion	Edema	**0.824**	0.469	0.374
	ET	**0.746**	0.389	0.326
	NCR†	**0.532**	0.461	0.302

on insertion, while the small-sample deletion comparison is inconclusive and we do not claim NCR superiority. Gradient saliency consistently outperforms Grad-CAM, so CCR's margin holds against the stronger baseline.

5.5 Qualitative explanation

Figure 6 shows the routing read-out for a representative case. The hard routing assignment (panel 4) reproduces the ground-truth and predicted segmentation *within the tumor*—a direct visual statement of routing = prediction—while the per-concept soft maps localize each clinical concept to its sub-region (NCR in the core, edema on the rim, ET in the enhancing ring). A clinician reads these maps directly as {NCR, Edema, ET}, which a gradient heatmap cannot provide. In a representative hard case (Fig. 8) the routing flags enhancing tumor that

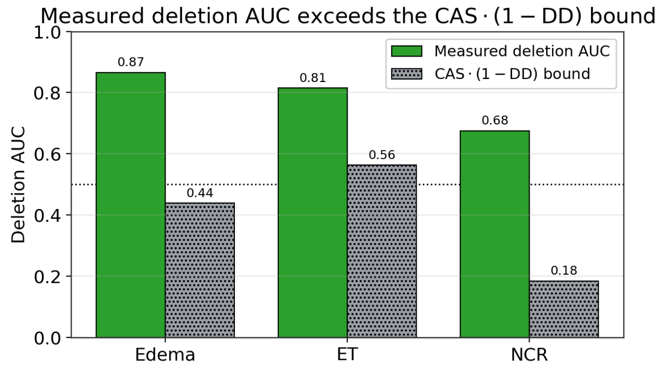

Fig. 4. The measured deletion AUC (green) exceeds the $\mathrm{CAS} \cdot (1 - \mathrm{DD})$ lower bound (gray) of Proposition 2 for every concept, confirming the bound empirically; dotted line marks the 0.5 reference.

Table 4. Ablations on the test split. A1 removes the alignment loss; A4 removes the warm-up phase. [Ablation runs in progress—numbers to be filled on completion.]

	Full CCR-Net	A1: no $\mathcal{L}_{\text{align}}$	A4: no warm-up
$\mathrm{CAS}_{\mathrm{fg}}$ (Edema)	0.935	[∼]	[∼]
$\mathrm{CAS}_{\mathrm{fg}}$ (ET)	0.931	[∼]	[∼]
$\mathrm{CAS}_{\mathrm{fg}}$ (NCR)	0.682	[∼]	[∼]
Dice (WT)	0.915	[∼]	[∼]
Max expert utilization (%)	[∼]	[∼]	[∼]

the decoder under-segments, showing that the routing explanation can carry diagnostic signal independent of—and more informative than—the final mask.

5.6 Ablations

We ablate the two design choices the theory predicts are essential. **A1 (no $\mathcal{L}_{\text{align}}$)** disables the alignment loss in every phase; **A4 (no warm-up)** activates alignment from epoch 1. Table 4 reports the outcome.

The hypotheses, which the theory and training dynamics (Fig. 7) both predict, are: A1 collapses $\mathrm{CAS}_{\mathrm{fg}}$ toward the random baseline while leaving Dice largely intact—showing that concept alignment is *learned* from $\mathcal{L}_{\text{align}}$, not an incidental by-product of segmentation; and A4 drives expert utilization to a single dominant expert (collapse), degrading both CAS and Dice and confirming that warm-up is necessary.

5.7 Training dynamics

Figure 7 traces $\mathrm{CAS}_{\mathrm{fg}}$ over training. Alignment is near zero during warm-up (routing unsupervised), rises sharply the moment $\mathcal{L}_{\text{align}}$ activates at epoch 11,

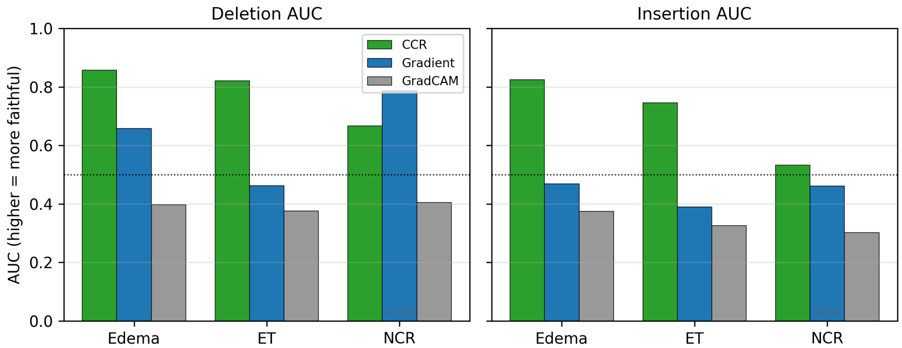

Fig. 5. Faithfulness on a matched set (deletion left, insertion right; higher is more faithful, Eq. 5). CCR (green) dominates gradient saliency and Grad-CAM on edema and ET; the dotted line marks the 0.5 reference. NCR is measured on n=16 cases.

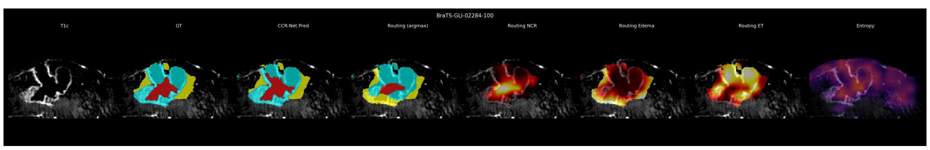

Fig. 6. Routing maps (case BraTS-GLI-02284). Left to right: T1ce, ground truth, prediction, hard routing assignment, soft routing for NCR / Edema / ET, and routing entropy. The argmax routing matches GT/prediction inside the tumor; each soft map localizes its concept; entropy peaks at concept boundaries.

and is held through refinement; edema and ET converge near 0.93 while NCR plateaus at the resolution-imposed ceiling. The curve is the temporal mirror of the A1 ablation: removing $\mathcal{L}_{\text{align}}$ removes exactly the rise visible at epoch 11.

6 Discussion and Limitations

Necrotic core resolution. At the 16^3 bottleneck a case contributes only tens of NCR tokens, capping CAS and faithfulness for that concept; this is the single consistent weak spot across all metrics. A finer bottleneck trades semantic context for resolution, and reconciling the two is open. **Calibration.** Routing entropy provides a free per-voxel uncertainty map, but its absolute calibration is poor (expected calibration error ≈ 0.73): peaked routing at low temperature is confident even where token-level accuracy is low. We therefore treat entropy as a qualitative uncertainty signal and make no calibration claim. **Soft training vs. hard inference.** During training all experts contribute via soft weighting; the faithfulness claim concerns inference-time hard dispatch, and our deletion/insertion analysis uses hard-routing maps accordingly (Sec. 5.4). **Concept granularity.** K=4 imposes hard concept boundaries; transition-zone tokens are

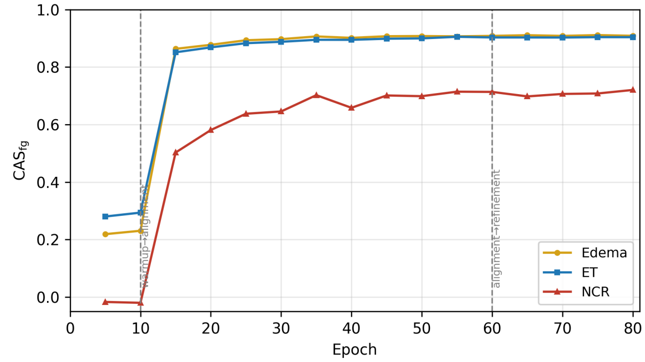

Fig. 7. CAS_{fg} across the three-phase curriculum (dashed lines: phase boundaries). Alignment emerges only after $\mathcal{L}_{align}$ activates at epoch 11, directly mirroring the A1 ablation.

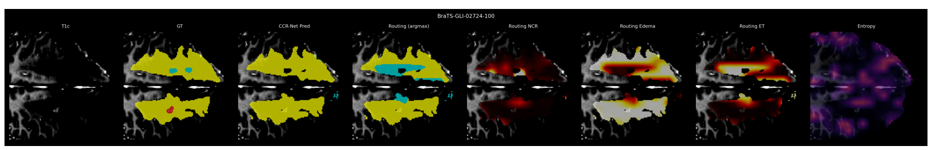

Fig. 8. Hard case (BraTS-GLI-02724): the routing assignment flags enhancing tumor that the final decoder mask under-segments, illustrating that the routing explanation can be more informative than the prediction itself.

forced into one concept, matching the BraTS protocol but discarding mixed-tissue nuance.

7 Conclusion

We introduced Clinical Concept Routing: making the routing decision at the segmentation bottleneck the faithful, clinically-labeled, per-voxel explanation, computed in one forward pass. We detailed the router, expert bank, dispatch, objective, and curriculum that make this work, and showed that CCR-Net's explanation is structurally faithful with alignment quality equal to the Concept Alignment Score. On an identical causal protocol the routing is far more faithful than Grad-CAM or gradient saliency on the concepts that admit reliable measurement. CCR is the first method to deliver intrinsic, concept-labeled, faithful explanation for 3D volumetric dense prediction; extending the principle to arbitrary backbones (Proposition 2) and to additional anatomies is promising future work.

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